

Accelerating Evidence-Grounded FAQ Assistants: A Distributed Agentic RAG Framework with Parallelized Graph-Retrieval and Multi-Level Citations

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1 Problem Statement

Advanced Retrieval-Augmented Generation (RAG) systems, particularly those utilizing Agentic Workflows or Knowledge Graphs, suffer from extreme inference latency due to sequential "retrieve-criticize-refine" loops. Furthermore, providing accurate, verifiable citations from diverse multi-source knowledge bases is computationally expensive. This makes real-time deployment for large-scale datasets, such as medical FAQ systems or legal databases, practically unfeasible using standard sequential processing.

2 Motivation

In high-stakes domains like healthcare, AI "hallucinations" are unacceptable. Users require evidence-based answers with verifiable citations. By applying Parallel and Distributed Computing (PDC) techniques—such as Distributed Task Scheduling and Parallel Indexing—we can implement sophisticated RAG pipelines that provide State-of-the-Art (SOTA) accuracy without the prohibitive time delays of traditional architectures. This research bridges the gap between high-fidelity NLP and high-performance computing.

3 Objectives

- **System Design:** Implement a 3-tier hierarchical Graph-RAG system for FAQ retrieval.
- **Latency Reduction:** Decrease end-to-end inference latency of the agentic self-correction loop by at least 50% using parallel task execution.
- **Citation Reliability:** Achieve a citation accuracy of $> 90\%$ by integrating automated parallel "Source Planning."
- **PDC Benchmarking:** Analyze the system's Speedup and Efficiency as the number of distributed worker nodes increases.

4 Technical Contributions

- **Parallelized U-Retrieval:** Implementation of a distributed "Top-down/Bottom-up" retrieval strategy where tag-indexing and graph-refinement occur across parallel threads [3].
- **Distributed Agentic Workflow:** Utilizing a task-parallel framework (Ray/Dask) to run "RAG-Critic" and "Executor" agents concurrently, eliminating the "stop-and-wait" bottleneck of serial feedback [1].
- **Scalable Ingestion:** A parallelized preprocessing pipeline for Triple Graph Construction, utilizing multi-GPU entity extraction to handle massive PDF corpora.

5 Dataset(s)

- **MedQA & PubMedQA:** For testing evidence-based medical FAQ reasoning and multi-hop retrieval.
- **MedOmniKB:** A multi-genre medical knowledge repository for complex source planning [2].
- **ASQA:** For evaluating long-form generation and citation precision.

6 Proposed Methodology / Flow

1. **Distributed Ingestion:** Partition raw corpora across nodes for parallel semantic chunking.
2. **Parallel Indexing:** Construct a Triple Graph (Entity-Definition-Source) stored in a distributed vector database.
3. **Concurrent Agentic Retrieval:** Use a Planner Agent to generate multiple parallel queries; Retrieval Workers execute "U-Retrieval" in parallel; a Parallel Critic evaluates evidence simultaneously.
4. **Refined Generation:** Synthesis of final grounded answers with multi-level citations.

7 Expected Outcomes

- **Performance:** Significant reduction in "Time-to-First-Token" compared to serial agentic RAG.
- **Quality:** A zero-hallucination system where every claim is grounded in a retrieved graph-node.
- **Scalability:** Quantitative evidence of linear speedup, proving the framework is ready for enterprise-scale knowledge management.

References

- [1] Dong et al. (2025). *RAG-Critic: Leveraging Automated Critic-Guided Agentic Workflow for Retrieval Augmented Generation*. ACL. <https://aclanthology.org/2025.acl-long.179/>
- [2] Chen et al. (2025). *Towards Omni-RAG: Comprehensive Retrieval-Augmented Generation for Large Language Models in Medical Applications*. ACL. <https://aclanthology.org/2025.acl-long.742/>
- [3] Wu et al. (2025). *Medical Graph RAG: Evidence-based Medical Large Language Graph Retrieval Augmented Generation*. ACL. <https://aclanthology.org/2025.acl-long.1381/>